cotyledons are thin. The portion of embryonal axis projecting beyond the cotyledons is called **radicle** or embryonic root. The other end of the axis called embryonic shoot is the **plumule**. Embryonal axis above the level of cotyledon is called **epicotyl** whereas the cylindrical region between the level of cotyledon is called **hypocotyl**(Figure 1.23 a).

***Oryza* seed (example for Monocot seed)**

The seed of paddy is one seeded and is called **Caryopsis**. Each seed remains enclosed by a brownish husk which consists of glumes arranged in two rows. The seed coat is a brownish, membranous layer closely adhered to the grain. Endosperm forms the bulk of the grain and is the storage tissue. It is separated from embryo by a definite layer called **epithelium**. The embryo is small and consists of one shield-shaped cotyledon known as **scutellum** present towards lateral side of embryonal axis. A short axis with plumule and radicle protected by the **root cap** is present. The plumule is surrounded by a protective sheath called **coleoptile**. The radicle including root cap is also covered by a protective sheath called **coleorhiza**. The scutellum supplies the growing embryo with food material absorbed from the endosperm with the help of the epithelium (Figure 1.23 b).

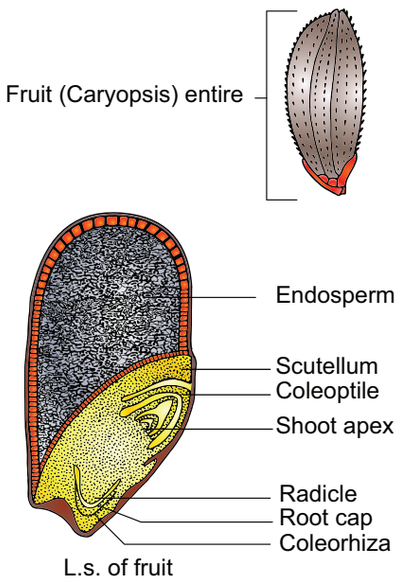



**Figure 1.23(b)** Monocot seed - *Oryza sativa*

**Activity**

Soak seeds of green gram for three hours. Drain the water and place few seeds in a clean tray containing moist cotton or filter paper. Allow the seeds to sprout. Collect the sprouted seeds, cut open and observe the parts. Record your observation.

## 1.7 Apomixis

Reproduction involving fertilization in flowering plants is called amphimixis and wherever reproduction does not involve union of male and female gametes is called apomixis.

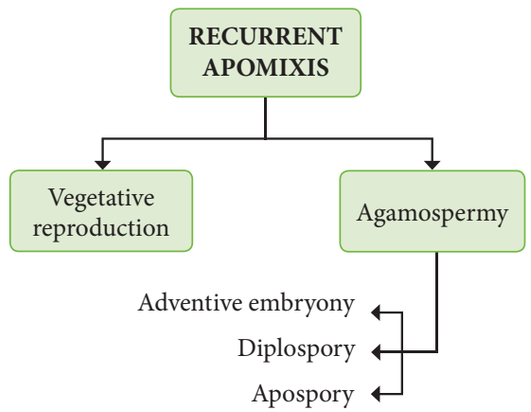
The term Apomixis was introduced by Winkler in the year 1908. It is defined as the substitution of the usual sexual system (Amphimixis) by a form of reproduction which does not involve meiosis and syngamy.

Maheswari (1950) classified Apomixis into two types - Recurrent and Non recurrent

**I. Recurrent apomixis:** It includes vegetative reproduction and agamospermy

**II. Non recurrent apomixis:** Haploid embryo sac developed after meiosis, develops into a embryo without fertilization.

**I.** The outline classification of Recurrent apomixis is given below.




**i) Vegetative reproduction:** Plants propagate by any part other than seeds



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